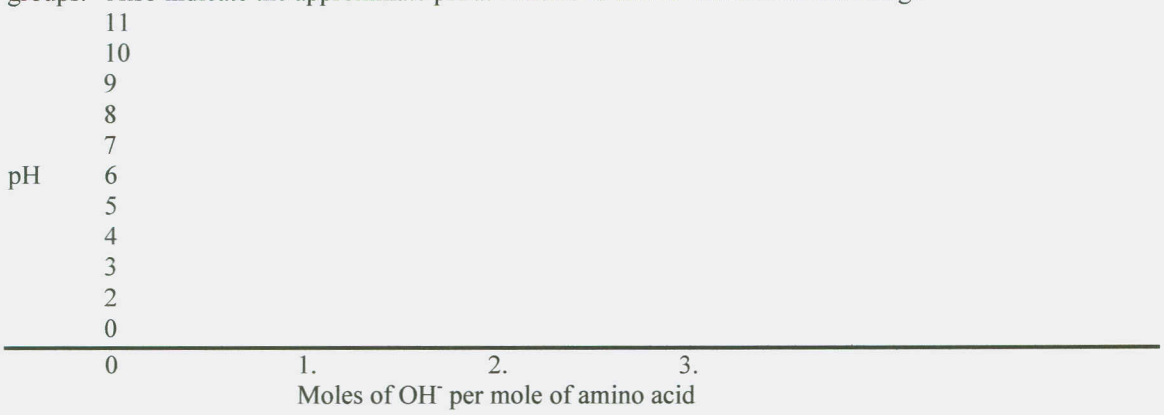
Biochemistry Cume February 20, 2009

1. (20 pts) Sketch a titration curve for the amino acid Gln and indicate the pKa values for all titratable groups. Also indicate the approximate pH at which this amino acid has no net charge.



2. (10 pts) Estimate the height of hemerythrin that is shown below. It is an alpha helical protein. (Helpful hints: 5.4 Å rise/turn, 3.6 residues/turn, 1.5 Å rise/residue).

_____ Å

3. (70 pts) Name molecules from the list below that fit each description. Answers can be used more than once, and some blanks require more than one answer. Choose the best answer(s).

List: Asn, Gln, Glu, Gly, Ile Lys, Met, Pro, Ser, Tyr, Val

A. A pair can form a salt bridge (electrostatic attraction) using their side-chains_____
B. The geometry of this **imino acid** is such that it does not fit into the alpha-helix_____
C. Non-polar, more atoms than Val, and find them in the center of a protein _____
D. Two chiral centers _____
E. The largest _____
F. The smallest _____
G. Electrically neutral polar side chains _____
H. Two residues that are most frequently found in a triple-helical collagen _____
I. An essential sulfer-containing _____
J. Beta-branched _____
K. Closely resembling (isosteric to) Cys _____
L. Hydroxyl group _____
M. Amide group _____
N. Sodium salt of this food additive is commonly known as MSG _____
O. Achiral _____
P. Isomer (same molecular formula) of Leu _____
Q. Aromatic ring _____